PTPC: Two-Stage Patch-Token Prototype Calibration for Few-Shot Object Detection

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Abstract

Few-shot object detection (FSOD) aims to recognize novel-class objects from only a handful of support examples. Strong CNN-based detectors localize these novel objects well but score them unreliably. Existing prototype-based recalibration addresses this by rescoring proposal crops against isolated per-class prototypes, but discards full-image context and offers no defense against mis-scoring base-class objects as novel. We propose an inference-time module, Two-Stage Patch-Token Prototype Calibration (PTPC), that uses a frozen foundation-model encoder to recalibrate the confidence scores of any CNN-based FSOD detector. In its first stage, PTPC encodes the query image once and pools patch tokens inside each detector box; the pooled descriptor is then scored against novel-class support prototypes to produce a recalibrated detection confidence. In the second stage, a Base-Confusability Guard suppresses detections whose crop-level features are closer to a cached base-class prototype than to any novel prototype. Detector weights, proposals, and class labels are unchanged throughout these two stages. PTPC improves detection accuracy across multiple CNN detector families on PASCAL VOC and MS COCO. Paired with MFDC, PTPC reaches mean VOC AP50 of 62.7, surpassing the strongest published CNN-based baseline by 2.0 points (60.7). On DeFRCN VOC 5-shot it roughly halves Expected Calibration Error and Brier score against the prior prototype-recalibration baseline. A controlled ablation shows that swapping in the foundation-model encoder alone is insufficient; calibration only recovers once PTPC's patch-token pooling and the Guard are added on top. PTPC thus brings foundation-model discrimination to deployed CNN-based FSOD systems without replacing the detection pipeline.

1 Introduction

Few-shot object detection (FSOD) must recognize novel object categories from a small support set while still localizing objects accurately in cluttered images. Most deployed FSOD systems inherit the CNN detector stack built around ResNet/Faster R-CNN/FPN-style proposal and classification heads [17, 18, 27, 31, 33, 35, 37, 42]. The recent shift toward transformers and foundation models has changed the accuracy frontier: ViT and DETR-style architectures [3, 4] and frozen-feature FSOD systems such as DE-ViT [50], FM-FSOD [13], and CD-ViTO [10] show that large pretrained representations carry powerful novel-class discriminative signal, but typically replace or substantially rework the detector, increasing memory, latency, or engineering commitment.

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CNN-based FSOD detectors often provide useful boxes, but their novel-class confidence scores are less reliable than their localization behavior suggests. Retraining the detector whenever the support set changes is costly: meta-learning methods [7, 15, 20, 36, 39, 45] and fine-tuning or distillation methods [31, 35, 37, 38, 42, 44] modify the detector weights, while full foundation-model pipelines change the deployment stack. We instead ask whether a fixed CNN detector can keep its efficient localization machinery while borrowing only the discriminative feature space of a frozen foundation model for confidence calibration.

The closest existing post-hoc answer is DeFRCN's Prototype Calibration Block (PCB) [31]. PCB has the right shape for our setting, but two failure modes limit it in the FSOD regime: its per-detection score is broken which is a context-blind global crop descriptor combined with a cross-class-blind isolated cosine, and it has no defense against base-class leakage. PTPC replaces PCB's calibration branch with a plug-and-play, two-stage inference-time module on a frozen foundation-model encoder; §3 formalizes the design.

Empirically, PTPC raises DeFRCN VOC 5-shot Split-1 AP50 from 56.09 (PCB baseline) to 67.58 (full ablation in Table 3), more than halving ECE and Brier; the three-split mean gains transfer across CNN detector families (FSCE, MFDC) and to MS COCO. A controlled ablation isolates the source of the improvement to PTPC's design rather than the encoder swap.

Contributions.

1. **Two failure modes of PCB-based FSOD calibration.** (i) PCB's per-detection scoring is doubly broken: its global crop-CLS descriptor is *context-blind* which discards the spatial layout of object evidence inside the detector box, and its isolated per-class cosine is *cross-class-blind* which provides no normalizer for absolute-similarity inflation. The two defects compound. (ii) PCB has no defense against base-class leakage. Confident detections overlapping base-class ground truth retain high novel-class confidence. Neither has been quantified in prior FSOD post-hoc work.

2. **PTPC, a two-stage inference-time calibration branch** for fixed CNN-based FSOD detectors, built on a frozen foundation-model encoder, with each stage targeting one failure mode above. *Stage I* attacks (i) on both prongs: a patch-token RoI descriptor pools the encoder's full-image patch grid *inside* each detector box with center-weighted Gaussian weights which fixes context-blindness, and softmax-competitive cosine scores novel prototypes against one another fixing cross-class-blindness. *Stage II* adds a Base-Confusability Guard that uses cached base prototypes as negative evidence to suppress (ii)'s leakage. The detector, its proposals, and its class labels are unchanged.

3. **PTPC exceeds strong CNN-based and ViT/foundation-model baselines on VOC, with consistent gains across detectors and on COCO.** Paired with strong CNN based detector, PTPC exceeds mean AP50 against most of the CNN and ViT based baselines. The improvement also transfers across different based detectors. To MS COCO it also improves over the prior CNN-based baselines at every evaluated shot count. PTPC achieves this at a fraction time compared to the most of the ViT based baselines.

2 Related Work

CNN-based few-shot object detection. FSOD systems built on the Faster R-CNN / FPN / Mask R-CNN lineage with ResNet backbones [17, 18, 27, 33] adapt these detectors to novel classes through three broad strategies. Meta-learning trains support-conditioned detectors or episodic learners [7, 15, 20, 36, 39, 45]. Refinement methods improve support usage through multi-scale positives, hallucinated examples, semantic relations, query adaptation, or retention mechanisms [9, 14, 19, 41, 49, 52]. Fine-tuning and decoupling methods adapt a base detector using class-balanced schedules, contrastive proposal objectives, distillation, prototype elaboration, or augmentation [2, 5, 16, 21, 22, 23, 24, 28, 29, 31, 35, 37, 38, 40, 42, 44, 48]. All three strategies modify detector weights; PTPC operates post-hoc on the detector's outputs.

Foundation-model FSOD detectors. A second line of work replaces the CNN detector with a transformer or foundation-model backbone. Open-vocabulary detectors [11, 25, 51] use language-aligned or image-level supervision to expand detector vocabularies. FS-DETR [1] adapts a transformer detector to novel classes without retraining; DE-ViT [50] combines frozen DINOv2 ViT-L/14 features with region propagation and background prototypes; FM-FSOD [13] pairs frozen DINOv2 features with language-model contextualization; CD-ViTO [10] extends foundation-model-first FSOD to cross-domain transfer through additional adaptation. PTPC uses the same representational signal only as a calibration branch on top of an existing CNN detector, preserving the deployed detection pipeline.

Post-hoc calibration and prototype scoring. Confidence calibration has been studied extensively in image classification [12] but less so for detection. Scalar post-hoc transformations such as temperature scaling adjust classifier logits after training but do not address class-level confusion: they reduce calibration error while leaving detection accuracy essentially unchanged. The most directly comparable prior method is PCB [31], with subsequent FSOD calibration work following the same template [8]. Prototype scoring itself traces to Prototypical Networks [34], and prototype interaction has been adapted for FSOD in training-time pipelines [7, 23, 28, 40, 43]. PTPC inherits this post-hoc prototype lineage; §3 details how it revises PCB.

3 Method

3.1 Problem Setting

Following the standard FSOD setup used in prior work [7, 20, 31, 37, 38, 43], object categories are partitioned into two sets C_b and C_n, where C_b is the base set with abundant annotations and C_n is the novel set with only K labeled support examples per class. Let D_{base} and D_{novel} denote the corresponding base-training and novel-support data. As in fine-tuning-based FSOD [31, 37, 38], a detector is first trained on D_{base} and then adapted to the few-shot regime using D_{novel}. At test time, the goal is to classify and localize objects in a query set D_{query}, whose categories belong to the FSOD label space induced by the base and novel classes.

3.2 PCB Revision

PCB [31] encodes each detection and support crop with a frozen ImageNet-pretrained ResNet-101, scores a detection against its predicted class prototype by isolated cosine similarity, and linearly fuses this score with the detector confidence. PTPC fixes the two failure modes via three design ingredients on a frozen foundation-model encoder substrate: patch-token RoI pooling and competitive novel-prototype scoring (jointly forming Stage I, §3.4, addressing failure mode (i)), and a Base-Confusability Guard that uses cached base prototypes as negative evidence (Stage II, §3.5, addressing failure mode (ii)). The encoder substrate (§3.3) is necessary but not sufficient—§4.3 shows that all three design ingredients are required. Figure 1 shows the full pipeline.

3.3 Shared Frozen Encoder

Detector outputs and calibration goal. Let D be the underlying CNN-based few-shot detector. For a query image $I \in [0,1]^{H_I \times W_I \times 3}$, where $H_I, W_I \in \mathbb{Z}^+$ are the image height and width, D returns a finite post-NMS detection set

$$\mathcal{Y}_D(I) = \{(b_i, c_i, s_i^d)\}_{i=1}^{N_I}, \tag{1}$$

where each b_i is an axis-aligned pixel-coordinate bounding box, $c_i \in C_n$ is the detector-predicted novel class, and $s_i^d \in [0,1]$ is the detector's confidence. PTPC keeps b_i and c_i unchanged, and only maps the scalar score through a calibration function g,

$$\hat{s}_i = g(b_i, c_i, s_i^d; I, D_{\text{novel}}, D_{\text{base}}) \in [0,1],$$

where $\hat{s}_i$ is the final recalibrated confidence used for ranking and evaluation.

Full-image patch-token encoding. PTPC uses a frozen foundation-model encoder ϕ whose parameters are fixed throughout calibration and whose full-image forward pass exposes a spatial grid of patch tokens. Let $S_\phi \in \mathbb{Z}^+$ denote the encoder's square full-image input resolution, and let $R(I) \in [0,1]^{S_\phi \times S_\phi \times 3}$ denote the query image after resizing to that encoder frame. The encoder maps the resized image to a grid of patch tokens,

$$\phi : [0,1]^{S_\phi \times S_\phi \times 3} \to \mathbb{R}^{H \times W \times d}, \qquad Z = \phi(R(I)), \tag{2}$$

where $H, W \in \mathbb{N}$ are the token-grid height and width, $d \in \mathbb{N}$ is the embedding dimension, and $Z_{u,v} \in \mathbb{R}^d$ is the token at grid row $u \in \{1,\dots,H\}$ and column $v \in \{1,\dots,W\}$. Equivalently, flattening the grid gives

$$T(I) = \{(p_r, t_r)\}_{r=1}^{HW}, \qquad t_r = Z_{u,v} \in \mathbb{R}^d, \qquad r = (u-1)W + v, \tag{3}$$

with patch-center coordinate

$$p_r = (x_r, y_r) \in [0, S_\phi]^2, \qquad x_r = \frac{v - \frac{1}{2}}{W} S_\phi, \qquad y_r = \frac{u - \frac{1}{2}}{H} S_\phi, \tag{4}$$

placing each token at the center of its grid cell in the resized image frame.

3.4 Stage I: Patch-Token Novel-Prototype Recalibration

Stage I corresponds to the upper branch of Figure 1.

Token selection inside each detector box. Define Π as the deterministic rescaling map from the original image frame to the $S_\phi \times S_\phi$ encoder frame:

$$\tilde{b}_i = \Pi(b_i) = (\tilde{x}_i^1, \tilde{y}_i^1, \tilde{x}_i^2, \tilde{y}_i^2), \qquad \tilde{x}_i^j = S_\phi \frac{x_i^j}{W_I}, \qquad \tilde{y}_i^j = S_\phi \frac{y_i^j}{H_I}, \quad j \in \{1,2\}, \tag{5}$$

with center

$$p_{b_i} = (x_{b_i}, y_{b_i}) = \left(\frac{\tilde{x}_i^1 + \tilde{x}_i^2}{2}, \frac{\tilde{y}_i^1 + \tilde{y}_i^2}{2} \right). \tag{6}$$

Using the token centers $p_r = (x_r, y_r)$ from Eq. (4), PTPC selects token indices by

$$\Omega(b_i) = \begin{cases} \{ r \in \{1, \dots, HW\} \mid \tilde{x}_i^1 \le x_r \le \tilde{x}_i^2,\ \tilde{y}_i^1 \le y_r \le \tilde{y}_i^2 \}, & \text{if this set is nonempty,} \\ \left\{ \arg\min_{r \in \{1,\dots,HW\}} \| p_r - p_{b_i} \|_2 \right\}, & \text{otherwise,} \end{cases} \tag{7}$$

falling back to the nearest token whenever no center lies inside the rescaled box so that $\Omega(b_i)$ is never empty.

Center-weighted RoI pooling. For each detector-predicted box b_i, PTPC pools over the nonempty selected token set $\Omega(b_i) \subseteq \{1, \dots, HW\}$. For the rescaled query box $\tilde{b}_i = (\tilde{x}_i^1, \tilde{y}_i^1, \tilde{x}_i^2, \tilde{y}_i^2) \in [0, S_\phi]^4$, define its encoder-frame width and height as

$$w_i = \tilde{x}_i^2 - \tilde{x}_i^1 \in (0, S_\phi], \qquad h_i = \tilde{y}_i^2 - \tilde{y}_i^1 \in (0, S_\phi]. \tag{8}$$

The token stride is a positive scalar $\delta \in (0, S_\phi]$; for a regular $H \times W$ token grid, $\delta = \min(S_\phi/W, S_\phi/H)$. PTPC sets the horizontal and vertical Gaussian bandwidths to

$$\sigma_{x,i} = \max\left(\frac{w_i}{2}, \delta \right), \qquad \sigma_{y,i} = \max\left(\frac{h_i}{2}, \delta \right), \qquad \sigma_{x,i}, \sigma_{y,i} \in [\delta, S_\phi/2]. \tag{9}$$

This makes the pooling scale follow the box size while the stride floor prevents collapse for tiny boxes. For a selected token $r \in \Omega(b_i)$, let $p_r = (x_r, y_r) \in [0, S_\phi]^2$ be its center and $p_{b_i} = (x_{b_i}, y_{b_i}) \in [0, S_\phi]^2$ be the box center. Its center distance is measured by the normalized squared distance

$$d_{r,i} = \left(\frac{x_r - x_{b_i}}{\sigma_{x,i}} \right)^2 + \left(\frac{y_r - y_{b_i}}{\sigma_{y,i}} \right)^2 \in [0, \infty), \tag{10}$$

and the corresponding unnormalized Gaussian weight is

$$\alpha_r(b_i) = \exp\left(-\frac{1}{2} d_{r,i} \right) \in (0, 1], \tag{11}$$

which we normalize over the selected token set:

$$a_r(b_i) = \frac{\alpha_r(b_i)}{\sum_{u \in \Omega(b_i)} \alpha_u(b_i)}. \tag{12}$$

With $\mathrm{norm}(z) = z/\|z\|_2$ denoting ℓ_2 normalization, the pooled patch-token RoI descriptor is

$$q_p(I, b_i) = \mathrm{norm}\left(\sum_{r \in \Omega(b_i)} a_r(b_i)\, \mathrm{norm}(t_r) \right) \in \mathbb{R}^d. \tag{13}$$

Tying $\sigma_{x,i}, \sigma_{y,i}$ to box size lets the pooling footprint track object scale while emphasizing tokens near the box center.

Novel patch supports and prototypes. For each novel class $c \in C_n$, let the K-shot support set be

$$\mathcal{S}_c = \{(I_c^j, b_c^j)\}_{j=1}^K. \tag{14}$$

Applying the same patch-token extractor (Eqs. (5)–(13)) to each support pair gives a patch descriptor

$$q_p(I_c^j, b_c^j) \in \mathbb{R}^d, \tag{15}$$

and averaging across the K shots produces the class patch prototype

$$\mu_c^p = \mathrm{norm}\left(\frac{1}{K} \sum_{j=1}^K q_p(I_c^j, b_c^j) \right). \tag{16}$$

Competitive scoring over novel classes. Given a query detection (b_i, c_i), PTPC compares the patch-level query descriptor with all novel patch prototypes by competitive scoring over the novel label space:

$$s_p(b_i, c) = \frac{\exp\left(q_p(I, b_i)^\top \mu_c^p / \tau\right)}{\sum_{c' \in C_n} \exp\left(q_p(I, b_i)^\top \mu_{c'}^p / \tau\right)}. \tag{17}$$

Each inner product is exactly cosine similarity since $q_p(I, b_i)$ and μ_c^p are unit-norm by construction (Eqs. (13), (16)). The softmax temperature $\tau \in (0, \infty)$ controls competition sharpness. PTPC evaluates Eq. (17) at the detector-predicted label c_i, giving the Stage I patch-prototype score $s_p(b_i, c_i)$.

Fusion with detector confidence. PTPC fuses this novel-prototype evidence with the original detector confidence:

$$s_{\mathrm{fuse}}(b_i) = w_d\, s_i^d + w_\phi\, s_p(b_i, c_i), \qquad w_d, w_\phi \in [0,1], \quad w_d + w_\phi = 1. \tag{18}$$

The frozen-encoder branch acts as a corrective signal driven by the novel support set, rather than a replacement detector.

3.5 Stage II: Base-Confusability Guard

Stage II corresponds to the lower branch of Figure 1. Competitive scoring over only novel prototypes can still give high confidence to a base-class object if the detector labels it as novel. Stage II therefore adds a second pass after Eq. (18) to handle this base/novel confusion. It operates in a complementary crop-level feature space: each detection and each support example is encoded as a single global descriptor on the same frozen encoder, which keeps base-class evidence compact across the much larger label set $|C_b|$ while remaining commensurable with the novel side.

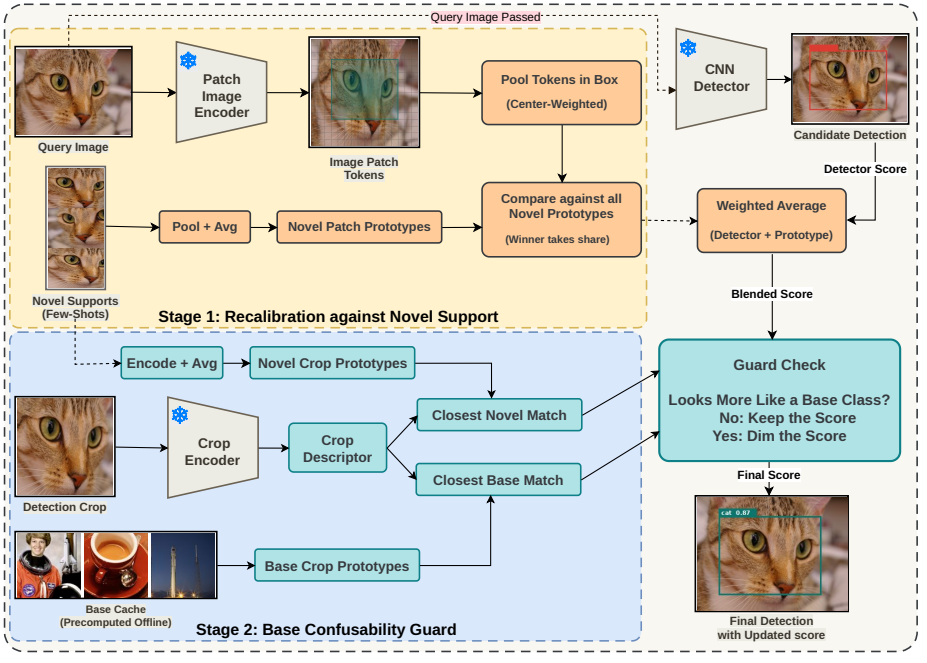


Figure 1: **Overview of PTPC.** A frozen foundation-model encoder with patch-token outputs runs once on the image. *Stage I* pools its patch tokens inside each detector box into a context-aware RoI descriptor, scores it against novel patch prototypes by softmax-competitive cosine similarity, and fuses with the detector confidence. *Stage II* encodes the same box as a crop-level descriptor and softly suppresses detections whose nearest cached base prototype is more similar than any novel prototype.

Crop-level guard descriptor. For each detector-predicted box b_i, let $\mathrm{crop}(I, b_i) \in [0,1]^{S\times S\times 3}$ denote the image region inside b_i, cropped from the query image I and resized to the encoder crop resolution $S \times S$, where $S \in \mathbb{Z}^+$. Let $\phi_{\mathrm{CLS}} : [0,1]^{S\times S\times 3} \to \mathbb{R}^d$ be the frozen encoder's global crop-token map; ϕ_{CLS} shares its weights with the patch encoder ϕ from §3.3 and differs only in input (a crop at resolution S) and output (the CLS embedding instead of the spatial patch grid). The crop-level descriptor is

$$q_g(I, b_i) = \mathrm{norm}\big(\phi_{\mathrm{CLS}}(\mathrm{crop}(I, b_i))\big) \in \mathbb{R}^d. \tag{19}$$

Novel crop supports and prototypes. Applying q_g to the novel support set $\mathcal{S}_c$ of Eq. (14) gives a crop-level support descriptor

$$q_g(I_c^j, b_c^j) \in \mathbb{R}^d, \tag{20}$$

and averaging across the K shots yields the novel crop prototype

$$\mu_c^g = \mathrm{norm}\left(\frac{1}{K}\sum_{j=1}^{K} q_g(I_c^j, b_c^j)\right). \tag{21}$$

Cached base prototypes. PTPC caches base-class prototypes using the same crop-level descriptor, but uses them only as negative evidence:

$$v_k^g = \mathrm{norm}\left(\frac{1}{M_k}\sum_{m=1}^{M_k} q_g(I_k^m, b_k^m)\right), \qquad k \in C_b. \tag{22}$$

Novel-versus-base similarity comparison. For each query detection, the guard compares the crop-level descriptor $q_g(I,b_i)$ with the nearest novel crop prototype and base crop prototype,

$$\delta_i^+ = \max_{c\in C_n} q_g(I,b_i)^\top \mu_c^g, \qquad \delta_i^- = \max_{k\in C_b} q_g(I,b_i)^\top v_k^g. \tag{23}$$

Soft confidence suppression. If the base similarity exceeds the novel similarity by margin m, PTPC does not relabel the detection as a base class. Instead, it softly suppresses the confidence by a multiplicative gate:

$$g_i = 1-(1-\gamma)\,\mathbf{1}\left[\delta_i^- - \delta_i^+ > m\right], \qquad s_{\mathrm{final}}(b_i) = g_i\, s_{\mathrm{fuse}}(b_i), \tag{24}$$

where m is the guard margin and $\gamma \in (0,1]$ is the suppression factor: when triggered, the gate multiplies the confidence by γ rather than relabeling or zeroing the detection, intervening only when the cached base manifold explains the detection better than any novel prototype.

4 Experiments

4.1 Experimental Setup

Datasets and Detectors. We use FSCE [35], DeFRCN [31], and MFDC [42] as CNN-based detector baselines, and DINOv2 [30], CLIP [32], and SigLIP [46] families as frozen foundation-model encoders. We evaluate on PASCAL VOC [6] and MS COCO [26] under the SNIDA evaluation protocol [38]. DeFRCN and MFDC baseline rows are rerun with the authors' official codebases; the small gap from their published means (DeFRCN: 49.6 vs. 51.9; MFDC: 57.7 vs. 59.0) is within the seed/hardware variance reported in independent reproductions, and all PTPC gains are measured against the same reimplemented baselines. For completeness, PTPC also exceeds the authors' published means.

Implementation Details. We deploy a single set of PTPC hyperparameters across all detectors and datasets to demonstrate calibrator-agnostic transfer; only the full-image resolution, crop resolution, and patch size vary with the foundation model.

The deployed Stage I branch uses only the patch-pooled prototype score; the crop-level descriptor is reserved for the guard. We fuse with $w_d = 0.4$ and $w_\phi = 0.6$ in Eq. (18), and use temperature $\tau = 0.1$ in Eq. (17). Base prototypes are computed from retained base-class crops, capped at 20 per base class; the guard uses margin $m = 0.05$ and suppression $\gamma = 0.3$. If a base cache is unavailable, the method falls back to Stage I without the guard.

Each support or detection crop is clipped to the image boundary and resized to $S\times S$. The DINOv2 variants use $S_\phi = 518$, $S = 224$, and patch size 14; CLIP-B/16 and SigLIP-B/16 use $S_\phi = S = 224$ with patch size 16; CLIP-L/14 uses $S_\phi = S = 224$ with patch size 14; SigLIP-L/16 uses $S_\phi = S = 256$ with patch size 16.

Table 1: PASCAL VOC comparison with recent CNN-based and ViT/foundation-model methods. We follow the full split/shot layout of SNIDA [33]: within each split block, columns 1/2/3/5/10 denote shots and each cell is AP50. Best and second-best are highlighted globally over all rows for each column.

Method	Novel Split 1					Novel Split 2					Novel Split 3					Mean
	1	2	3	5	10	1	2	3	5	10	1	2	3	5	10	
ICPE [28], AAAI 23	54.3	59.5	62.4	65.7	66.2	33.5	40.1	48.7	51.7	52.5	50.9	63.1	55.3	60.6	60.1	55.0
VFA [16], AAAI 23	57.7	64.6	64.7	67.2	67.4	41.4	46.2	51.1	51.8	51.6	48.9	54.8	56.6	59.0	58.9	56.1
s-ADP [6], ICCV 23	52.3	55.5	63.1	65.9	66.7	42.7	45.8	48.7	54.8	56.3	47.8	51.8	56.8	60.3	62.4	55.4
Norm-VAE [42], CVPR 23	62.1	64.9	67.8	69.2	67.5	39.9	46.8	54.4	54.2	53.6	58.2	60.3	61.0	64.0	65.5	59.1
DeFRCN [31], ICCV 21 (our impl.)	48.9	56.3	54.5	60.4	56.1	28.4	39.5	47.5	52.1	47.0	43.7	47.2	51.2	54.6	56.9	49.6
MFDC [27], ECCV 22 (our impl.)	61.7	66.1	60.9	68.8	66.5	41.3	41.1	45.5	54.8	52.9	58.0	60.7	59.1	63.0	64.5	57.7
SNIDA-MFDC [33], CVPR 24	64.9	67.9	69.7	71.4	70.5	42.2	**47.8**	54.5	56.6	54.9	58.1	61.3	60.7	63.6	66.0	60.7
ViT / Foundation-Model methods																
FS-DETR [1], ICCV 23	45.0	48.5	51.5	52.7	56.1	37.3	41.3	43.4	46.6	49.0	43.8	47.1	50.6	52.1	56.9	48.1
Meta-DETR [47], TPAMI 22	40.6	51.4	58.0	59.2	63.6	37.0	36.6	43.7	49.1	54.6	41.6	45.9	52.7	58.9	60.6	50.2
DE-ViT-B [50], CoRL 24	56.9	61.8	68.0	73.9	72.8	**45.3**	47.3	**58.2**	59.8	60.6	58.6	62.3	62.7	64.6	67.8	61.4
DE-ViT-L [50], CoRL 24	55.4	56.1	68.1	70.9	71.9	43.0	39.3	58.1	**61.6**	63.1	58.2	64.0	61.3	64.2	67.3	60.2
FM-FSOD-B [13], CVPR 24	40.9	52.8	59.5	68.3	71.4	33.5	36.1	48.1	53.6	59.3	41.9	52.6	54.9	62.8	**68.2**	53.6
FM-FSOD-L [13], CVPR 24	40.1	53.5	57.0	68.6	72.0	33.1	36.3	48.8	54.8	**64.7**	39.2	50.2	55.7	63.4	68.1	53.7
Ours (DeFRCN + PTPC-B)	63.6	68.3	70.3	73.2	71.5	36.4	43.4	58.1	59.8	60.2	55.8	58.3	61.6	65.2	65.6	60.8
Ours (DeFRCN + PTPC-L)	62.8	68.6	70.0	73.3	71.8	35.3	43.6	57.0	59.6	59.3	54.1	57.5	60.9	64.9	65.6	60.3
Ours (MFDC + PTPC-B)	68.7	**71.1**	71.1	74.8	73.6	43.2	41.5	51.3	60.3	59.9	**59.9**	**64.9**	**64.8**	**67.6**	67.8	**62.7**
Ours (MFDC + PTPC-L)	**69.1**	68.4	**73.0**	**76.2**	**76.9**	40.1	43.3	55.4	58.5	58.9	56.8	64.4	64.2	67.2	67.4	**62.7**

Metrics. We report standard detection metrics (AP, AP50, AP75) and calibration metrics: Expected Calibration Error [12] with 15 equal-width bins, and Brier score.

4.2 Main Results

PTPC is paired with DeFRCN and MFDC throughout the main comparisons, in two foundation-model variants: PTPC-B (DINOv2-B/14) and PTPC-L (DINOv2-L/14). The smaller PTPC-S (DINOv2-S/14) is excluded from these tables because its gains are smaller than PTPC-B and PTPC-L; its Split 1 numbers are reported in the foundation-model ablation (Table 5).

PASCAL VOC. On PASCAL VOC (Table 1), MFDC+PTPC reaches a mean AP50 of 62.7 on the SNIDA layout [33], surpassing both the strongest published CNN-based baseline (SNIDA-MFDC: 60.7) and the strongest ViT / foundation-model baseline (DE-ViT-B: 61.4). The two MFDC+PTPC variants tie on mean (62.7) but differ in shape: PTPC-L dominates Novel Split 1 (4 best cells) while PTPC-B leads Novel Split 3 (4 best cells); together the MFDC+PTPC variants occupy 9 of 15 split×shot best positions and 9 second-best positions. DeFRCN+PTPC-B reaches 60.8 mean, edging past SNIDA-MFDC under the same protocol.

MS COCO. On MS COCO (Table 2), MFDC+PTPC-B surpasses the strongest published CNN-based baseline (SNIDA-MFDC) at every shot count (1/2/3/5/10/30) and on mean AP under the SNIDA layout; MFDC+PTPC-L trails PTPC-B at every shot count (mean 18.4 vs. 18.8), indicating no benefit from encoder scaling beyond DINOv2-B on COCO. DeFRCN+PTPC-B also improves over SNIDA-MFDC at 2-/3-/5-/10-/30-shot and on mean AP, trailing only at 1-shot.

Table 2: MS COCO comparison with recent CNN-based and ViT/foundation-model methods. We follow the full shot-grid layout of SNIDA [53]; columns 1/2/3/5/10/30 denote shots and each cell is AP. Best and second-best are highlighted globally over all rows for each column. "–" marks shots not reported in the source paper.

Method	1	2	3	5	10	30	Mean
DeFRCN [31], ICCV 21	9.3	12.9	14.8	16.1	18.5	22.6	15.7
MFDC [42], ECCV 22	10.8	13.9	15.0	16.4	19.4	22.7	16.4
D&R [24], AAAI 23	8.3	12.7	14.3	16.4	18.7	21.8	15.4
Norm-VAE [44], CVPR 23	9.5	13.7	14.3	15.9	18.7	22.5	15.8
SNIDA-MFDC [53], CVPR 24	<u>12.0</u>	15.4	16.4	17.8	20.7	23.8	17.7
ViT / Foundation-Model methods							
Meta-DETR [47], TPAMI 22	7.5	–	13.5	15.4	19.0	22.2	–
FS-DETR [1], ICCV 23	7.0	8.9	10.0	10.9	11.3	–	–
DE-ViT-L [50], CoRL 24	–	–	–	–	**34.0**	<u>34.0</u>	–
FM-FSOD-L [13], CVPR 24	5.7	11.0	15.7	**21.9**	<u>27.7</u>	**37.0**	**19.8**
Ours (DeFRCN + PTPC-B)	10.4	15.6	<u>17.2</u>	18.4	20.8	24.3	17.8
Ours (DeFRCN + PTPC-L)	9.8	15.0	16.8	18.2	20.8	24.8	17.6
Ours (MFDC + PTPC-B)	**12.3**	**17.0**	**17.5**	<u>19.5</u>	21.6	24.6	<u>18.8</u>
Ours (MFDC + PTPC-L)	11.8	<u>16.2</u>	17.1	19.3	21.2	24.5	18.4

Inference cost. On a single NVIDIA RTX 3090, MFDC+PTPC-B adds 70–130 ms per image over the vanilla MFDC detector (VOC 0.14 → 0.21 s/img; COCO 0.16 → 0.29 s/img). For context, full-pipeline foundation-model detectors invoke the encoder per region proposal rather than once per image: DE-ViT [50] reports 0.22–0.83 s/img and FM-FSOD [13] ∼2.4 s/img per A100.

4.3 Ablations and Analysis

Ablations use DeFRCN with DINOv2-B unless noted; the foundation-model impact study uses MFDC.

Design-Space Ablation. Table 3 disentangles PTPC's contribution along three design axes on top of the frozen-encoder substrate: RoI pooling (crop-CLS vs. patch-RoI), novel-class competition (Comp., Eq. (17); off = PCB's isolated per-class cosine), and the Base-Confusability Guard (Eq. (24)), with the detector, support set, and evaluation protocol held fixed. The encoder (R101 vs. DINOv2-B) is varied as a control to isolate design from substrate.

A stronger frozen encoder alone is not enough. Under PCB's original crop-CLS + isolated-cosine recipe, swapping ResNet-101 for DINOv2-B gains 4.6 AP50 (56.09 → 60.68) but *worsens* ECE (0.217 → 0.369) and Brier (0.084 → 0.158). The two metrics are not in conflict: AP50 only sees per-class ranking, and DINOv2's sharper feature space ranks true positives above false positives more often, so AP50 rises; meanwhile isolated cosine has no cross-class normalizer, so the absolute score inflation produced by a sharper encoder spreads to wrong-class detections too, blowing out calibration.

Calibration recovers once PTPC's design choices are added. With both Comp. and Guard turned on, switching crop-CLS → patch-RoI adds 4.6 AP50 and roughly halves both ECE (0.205 → 0.114) and Brier (0.073 → 0.043). Within the patch-RoI block the Guard is the dominant single lever (+4.3 AP50 over Patch-RoI alone), while Comp. is a smaller

corrective (+0.6 AP50 without the Guard, +2.2 with it). Critically, adding the Guard alone to PCB's own ResNet-101 encoder (R101 + Crop-CLS + Guard) already nearly matches the full PTPC's calibration (ECE 0.217 → 0.114; Brier 0.084 → 0.044 vs. full PTPC's 0.043), isolating the calibration gain to the Guard design rather than the foundation-model encoder; AP50 in this configuration rises by only 1.8, however, so accurate detection requires the foundation-model encoder and patch-RoI pooling in addition. The full PTPC configuration is the only row that is simultaneously best in AP50 and best in calibration, confirming that both come from the design, not the encoder swap alone.

Table 3: Unified design-space ablation on DeFRCN VOC 5-shot. All rows share the same detector checkpoint, support set, and evaluation protocol; only the calibration branch changes. Splits 1–3 report novel-class AP50; Mean AP50 averages over the three splits. Best in each column is highlighted globally.

Encoder	Pooling	Comp.	Guard	Split 1	Split 2	Split 3	Mean AP50	ECE↓	Brier↓
—	—	—	—	49.98	52.68	50.31	50.99	0.151	0.059
R101	Crop-CLS	—	—	60.59	52.96	54.74	56.09	0.217	0.084
R101	Crop-CLS	✓	—	60.20	48.57	54.59	54.45	0.228	0.103
R101	Crop-CLS	—	✓	61.90	54.01	57.68	57.86	**0.114**	0.044
R101	Crop-CLS	✓	✓	62.04	51.63	57.90	57.19	0.123	0.053
DINOv2-B	Crop-CLS	—	—	65.06	56.54	60.43	60.68	0.369	0.158
DINOv2-B	Crop-CLS	✓	—	68.80	59.51	61.61	63.31	0.205	0.073
DINOv2-B	Crop-CLS	—	✓	70.23	59.76	63.75	64.58	0.166	0.054
DINOv2-B	Crop-CLS	✓	✓	68.94	58.49	61.46	62.96	0.205	0.073
DINOv2-B	Patch-RoI	—	—	65.95	56.77	60.82	61.14	0.381	0.168
DINOv2-B	Patch-RoI	✓	—	68.05	57.26	60.00	61.77	0.240	0.102
DINOv2-B	Patch-RoI	—	✓	72.11	59.05	65.14	65.43	0.173	0.058
DINOv2-B	Patch-RoI	✓	✓	**74.82**	**60.31**	**67.60**	**67.58**	**0.114**	**0.043**

Detector-Family Transfer. Figure 2 tests whether PTPC transfers across CNN-based FSOD detector families. PTPC raises novel-class AP50 for DeFRCN, FSCE, and MFDC at every shot count, with absolute gains tracking the baseline's headroom.

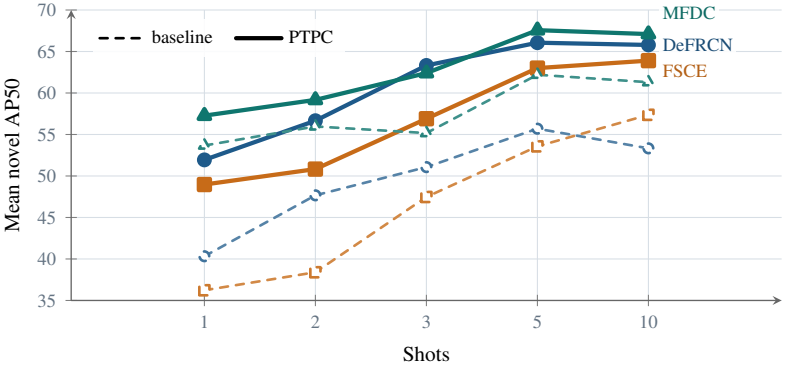
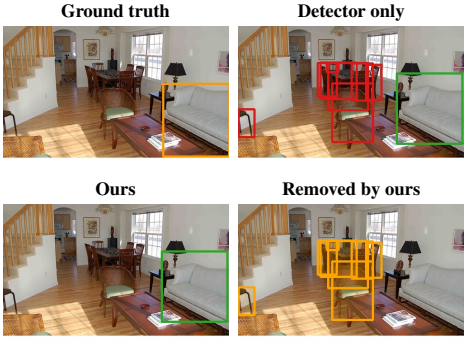
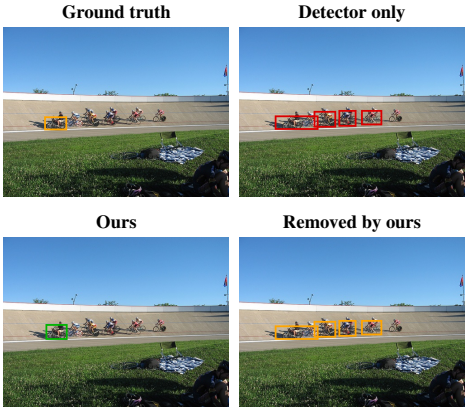


Figure 2: Line chart of transfer across detector families on PASCAL VOC. Each point is mean novel-class AP50 over novel splits 1–3 at the corresponding shot count. Solid lines denote PTPC-B (DINOv2-B encoder) and dashed lines denote the detector baselines.



(a) Image 000314: detector-only retains the visible true positive but produces eight false positives; ours keeps it and removes all eight.



(b) Image 000111: detector-only ranks a true positive below its top predictions and surfaces four false positives; ours promotes it and removes the flood.

Figure 3: **Qualitative guard behavior under cleanup and rank promotion.** The first case shows full cleanup around a retained true positive; the second shows promotion of a low-ranked true positive that detector-only does not display in its top predictions. The lower-right view in each case shows only detector-only predictions removed by our method.

Guard Selectivity. Table 4 tests whether the guard suppresses true positives alongside false positives. We log every post-NMS detection in the guard's intervention band $s \in (0.05, 1.0]$ (111,258 detections aggregated over the three splits) and bucket each by its maximum IoU with the in-image GT boxes (threshold ≥ 0.5, matching VOC): base GT, novel GT, or none. Under novel-class evaluation, a detection that overlaps a base GT is necessarily a false positive (the base object is being misread as a novel class), so a *high* base-GT suppression rate is desirable; a detection that overlaps a novel GT is a true positive at the candidate stage, so a *low* novel-GT suppression rate is desirable. The two rates therefore measure precision recovery and recall preservation respectively, and an ideal guard maximizes the first while minimizing the second.

Table 4: Guard-selectivity diagnostic on DeFRCN VOC 5-shot (aggregate over three splits) under the final patch-RoI-only Stage I design.

Overlap bucket	Checked	Suppressed	Suppression rate
Base GT overlap	78,311	66,374	84.8%
Novel GT overlap	6,464	300	4.6%
Background / no GT overlap	26,483	18,003	68.0%

The guard is targeted: it suppresses 84.8% of base-overlapping detections which is its intended failure mode, while leaving 95.4% of novel-overlapping detections intact, a roughly 18× selectivity ratio. Background detections sit between (68.0% suppressed), consistent with the guard catching unsupported high-confidence novel predictions that embed near a base prototype in cluttered scenes. Figure 3 confirms the same behavior visually.

Foundation-Model Impact. Table 5 asks whether PTPC's gains are tied to a specific frozen encoder. We hold the detector, proposals, and calibration hyperparameters fixed, and swap only the frozen encoder under the patch-RoI-only Stage I setting. To keep the sweep tractable, we report Novel Split 1 only because of resource constraints.

Every tested foundation-model encoder lifts the MFDC baseline on Split 1. Within Split 1, DINOv2-L/14 ranks best (shot-mean 72.70, winning 1/3/5/10-shot) and DINOv2-B/14 leads at 2-shot. So single split rankings do not extrapolate uniformly to the other splits. The 2-shot column is the least stable: all four image-text contrastive encoders (CLIP-B/L, SigLIP-B/L) fall below the MFDC baseline of 66.10—SigLIP-L/16 the worst at 54.11—suggesting that their text-aligned global features are less robust than DINOv2's dense self-supervised features under noisy few-shot prototype estimation.

Table 5: Frozen foundation-model ablation on MFDC. Results are Split 1 novel-class AP50 at 1/2/3/5/10 shots; Mean averages over the five shot settings.

Foundation model	1	2	3	5	10	Mean
–	61.70	66.10	60.90	68.80	66.50	64.80
DINOv2 [30]						
DINOv2-S/14	67.27	61.43	71.27	74.88	74.97	69.96
DINOv2-B/14	68.70	71.10	71.10	74.80	73.60	71.86
DINOv2-L/14	69.06	68.37	72.97	76.19	76.93	72.70
DINOv2-G/14	68.45	59.81	70.97	74.78	75.64	69.93
CLIP [32]						
CLIP-B/16	65.69	59.16	68.06	71.78	71.00	67.14
CLIP-L/14	64.29	62.99	68.20	74.86	72.94	68.66
SigLIP [46]						
SigLIP-B/16	59.95	63.58	68.29	74.53	75.40	68.35
SigLIP-L/16	59.44	54.11	66.05	72.67	74.66	65.39

Hyperparameter Sensitivity. Table 6 sweeps PTPC's three calibration hyperparameters: competitive-scoring temperature τ (Eq. (17)), guard margin m, and suppression factor γ (Eq. (24)) around their deployed values, holding the other two fixed in each sweep. The deployed values τ=0.1, m=0.05, γ=0.3 were tuned on Split 1; Splits 2–3 therefore measure transfer, not tuning. Mean AP50 varies by under 3.6 across the widest sweep (τ), and the deployed values jointly achieve the highest mean in every sweep, indicating that PTPC is not knife-edge sensitive to any single hyperparameter. This is a property of the calibrator's design rather than careful tuning.

Table 6: Hyperparameter sensitivity on DeFRCN VOC 5-shot. Each sub-table sweeps one of (a) τ (Eq. (17)), (b) m, or (c) γ (both Eq. (24)), with the other two held at the deployed value (highlighted). S1–S3 are split-wise novel AP50.

τ	S1	S2	S3	Mean
0.05	71.06	60.17	64.72	65.32
0.10	74.82	60.31	67.60	67.58
0.20	73.64	60.13	65.34	66.37
0.50	72.37	59.55	65.02	65.64
1.00	70.48	57.90	63.68	64.02

(a) Temperature τ

m	S1	S2	S3	Mean
0.000	71.85	60.33	64.40	65.53
0.025	72.62	60.21	64.94	65.92
0.050	74.82	60.31	67.60	67.58
0.100	73.01	60.43	65.34	66.26
0.150	72.20	60.02	64.89	65.70

(b) Guard margin m

γ	S1	S2	S3	Mean
0.10	72.83	59.85	65.19	65.96
0.20	73.00	60.00	65.21	66.07
0.30	74.82	60.31	67.60	67.58
0.50	73.24	60.31	65.20	66.25
0.70	73.52	60.43	65.17	66.37

(c) Suppression γ

5 Conclusion

PTPC ports foundation-model discrimination into deployed CNN-based FSOD detectors as a plug-and-play, inference-time calibration branch built on two coupled diagnoses of PCB: a doubly-broken per-detection score (context-blind descriptor, cross-class-blind cosine) and an unguarded base→novel leakage. A controlled design-space ablation isolates the headline gain to these design choices rather than to the encoder swap alone. The result is a calibration branch that recovers both AP50 and calibration on PASCAL VOC and MS COCO without touching the detector, its proposals, or its class labels. Code and hyperparameter configurations will be released upon acceptance.

Limitations. PTPC's gains depend on support-prototype quality, which can degrade when only a handful of examples are available per novel class; learned prototype aggregation or principled support-set augmentation are natural extensions, though absolute AP50 in the very-low-shot regime (1–3 shots) is already state-of-the-art among CNN-based methods. More fundamentally, PTPC is bounded by the underlying detector's post-NMS box recall: novel objects missed at detection cannot be recovered, since PTPC only rewrites the confidence of boxes the detector already produces. Full-pipeline foundation-model detectors [13, 50] do not share this constraint, at the cost of higher inference latency and a heavier deployment stack. The Base-Confusability Guard also targets base→novel leakage but does not explicitly address inter-novel confusion, which is left to Stage I's prototype competition. Finally, we evaluate on PASCAL VOC and MS COCO only; cross-domain transfer in the spirit of CD-ViTO [10] (e.g., Aquarium, ArTaxOr) is a natural next step.

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